

Elaina Lutz

(484) 681-0820 | elaina.lutz@temple.edu | www.linkedin.com/in/elaina-lutz | U.S. Citizen

EDUCATION

Temple University – College of Engineering

Bachelor of Science

Philadelphia, PA

Expected Graduation: May 2028

Major: Electrical and Computer Engineering, Electrical Concentration | **GPA:** 3.99/4.00

Honors: Dean's List: Fall 2024, Spring 2025, Fall 2025, Spring 2026 / Provost's Tuition Scholarship / Honors Program

Relevant Coursework: Power Electronics, Digital Circuit Design, Signals and Systems, Computation II

WORK EXPERIENCE

Neural Engineering Data Consortium

Software Engineer

Philadelphia, PA

August 2026 – Present

- Develop machine learning tools for biofilm image analysis, build automated training pipelines and biostatistics programs, create turn-key scripts for model training, and architect web-based diagnostic application.

Temple University College of Engineering

Engineering Computation I Peer Mentor

Philadelphia, PA

December 2025 – Present

- Evaluate and grade student submissions for code correctness, algorithmic efficiency, and analytical rigor, providing structured feedback on testable code practices and computational problem-solving.

Introduction to Engineering and Engineering Technology Teaching Assistant

August 2025 – December 2025

- Assisted first-year engineering students with technical problem solving and assignment verification, conducting weekly office hours providing guidance on systems-based design thinking and engineering fundamentals.

Constellation Energy – Crane Clean Energy Center

Programs & CMO Electrical Engineering Intern

Middletown, PA

June 2026 – August 2026

- Led verification workflow for breaker refurbishment and replacement operations, coordinating across site departments to ensure compliance with IEEE/NERC standards and documenting findings in technical reports.

- Assisted with aging management programs to ensure regulatory compliance and equipment reliability.

Strategic Electrical Engineering Intern

June 2025 – August 2025

- Performed inverter ID verification and asset management, validating equipment prior to operational restart.
- Conducted plant walkdowns to assess equipment readiness and analyzed electrical loading data for safe plant operations.

PROJECTS

Arcade Game Engine

April 2026

- Built a distributed game platform with Python and C++ servers communicating via TCP/JSON, featuring custom data structures optimized for performance and memory efficiency.
- Developed Pygame client with real time chat and game browsing UI, engineered polymorphic serialization for flexible wire formats, debugged complex network communication, and player state synchronization across remote connections.

Temperature Detector Circuit

December 2025

- Designed and implemented an analog temperature detection circuit using a thermistor, MOSFET, potentiometer, and LED to indicate an increase in temperature.
- Simulated circuit performance in NI Multisim and validated results through hardware testing and calibration.

Wind Turbine Analysis

June 2025

- Developed MATLAB and Excel simulation models to evaluate wind turbine efficiency under varying environmental conditions.

LEADERSHIP & ACTIVITIES

Eta Kappa Nu (HKN) – IEEE Honor Society

April 2026 - Present

Power and Energy Society (PES), IEEE Branch

Philadelphia, PA

Marketing Director

March 2026 – Present

Women in Engineering (WIE), IEEE Branch

Philadelphia, PA

Marketing Chair

Sept 2025 – Present

Theta Tau – Professional Engineering Fraternity

Philadelphia, PA

Treasurer

January 2026 – Present

Institute of Electrical and Electronics Engineers

Philadelphia, PA

Member

Sept 2024 – Present

SKILLS

Software: Python, C/C++, MATLAB, Linux/Unix Command Line, NumPy, NI Multisim, KiCad, AutoCAD, GitHub, Bash, Inventor 2025, Matplotlib, AWS, VS Code, EMACS, draw.io, Scopy, Slurm, Microsoft Office Suite

Hardware: Circuit Design & Breadboard Prototyping, Embedded Systems (Arduino, Micro:bit), Test & Measurement (Oscilloscopes, Multimeters), Soldering & PCB Assembly